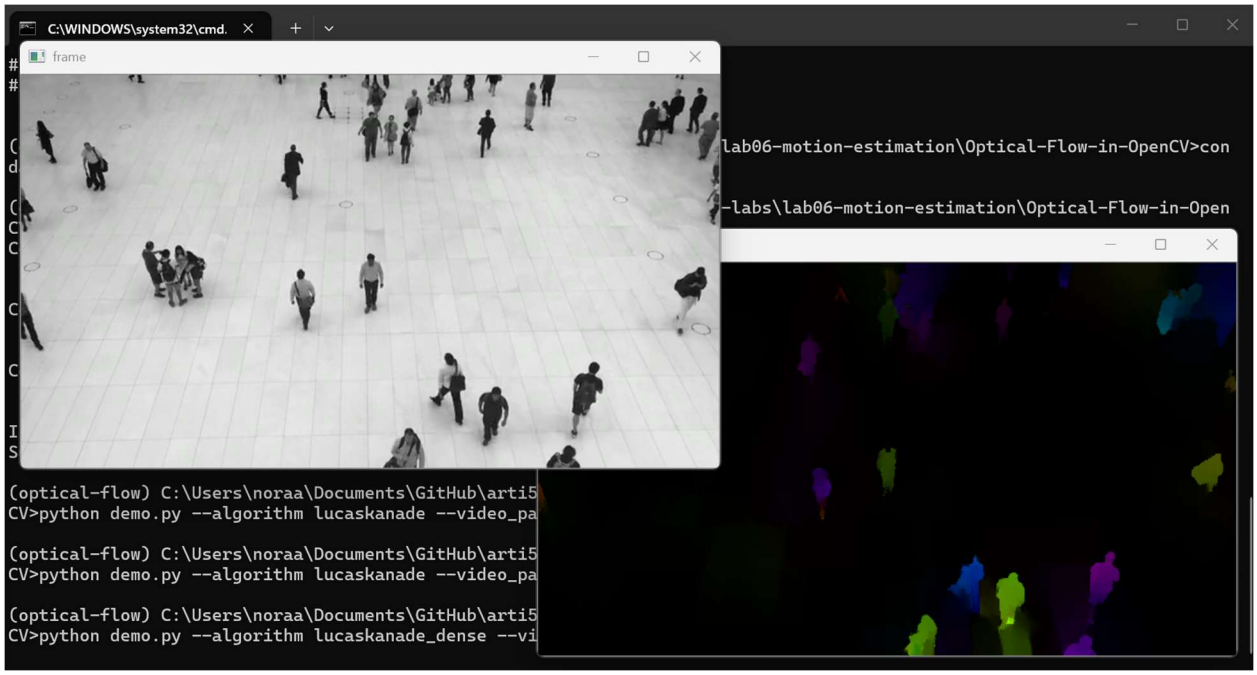
```
(cv_lab) C:\Users\noraa>cd "C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical
-Flow-in-OpenCV"

(cv_lab) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-OpenCV>con
da activate optical-flow

EnvironmentNameNotFound: Could not find conda environment: optical-flow
You can list all discoverable environments with `conda info --envs`.
```

```
C:\WINDOWS\system32\cmd.

#
#     $ conda activate optical-flow
#
# To deactivate an active environment, use
#
#     $ conda deactivate


(cv_lab) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-OpenCV>con
da activate optical-flow

(optical-flow) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-Open
CV>pip install -r requirements.txt
Collecting numpy==1.18.3
  Downloading numpy-1.18.3-cp37-cp37m-win_amd64.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 29.7 MB/s eta 0:00:00
Collecting opencv-python==4.2.0.34
  Downloading opencv_python-4.2.0.34-cp37-cp37m-win_amd64.whl (33.0 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 33.0/33.0 MB 13.9 MB/s eta 0:00:00
Collecting opencv-contrib-python==4.2.0.34
  Downloading opencv_contrib_python-4.2.0.34-cp37-cp37m-win_amd64.whl (39.5 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 39.5/39.5 MB 26.2 MB/s eta 0:00:00
Installing collected packages: numpy, opencv-python, opencv-contrib-python
Successfully installed numpy-1.18.3 opencv-contrib-python-4.2.0.34 opencv-python-4.2.0.34

(optical-flow) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-Open
CV>python demo.py --algorithm lucaskanade --video_path videos/crowd.mp4

(optical-flow) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-Open
CV>
```

Lucas-Kanade Optical Flow Result :

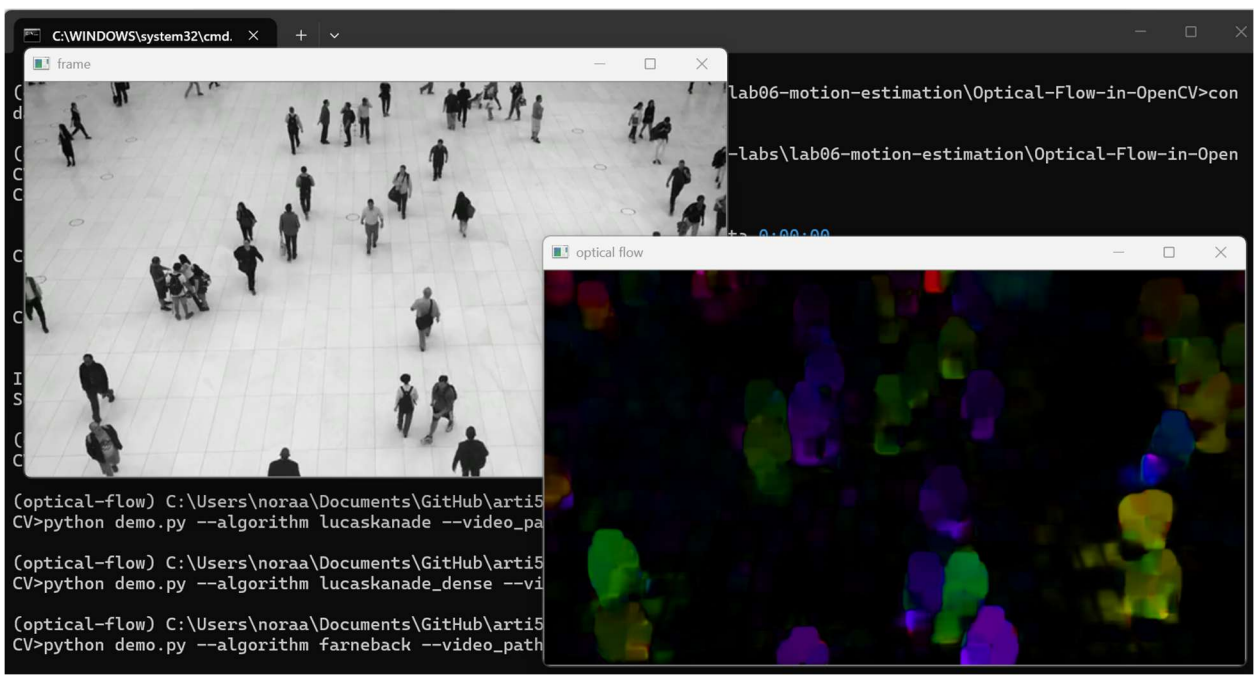
Dense Lucas-Kanade Result:

```
(optical-flow) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-Open
CV>python demo.py --algorithm lucaskanade_dense --video_path videos/crowd.mp4
```



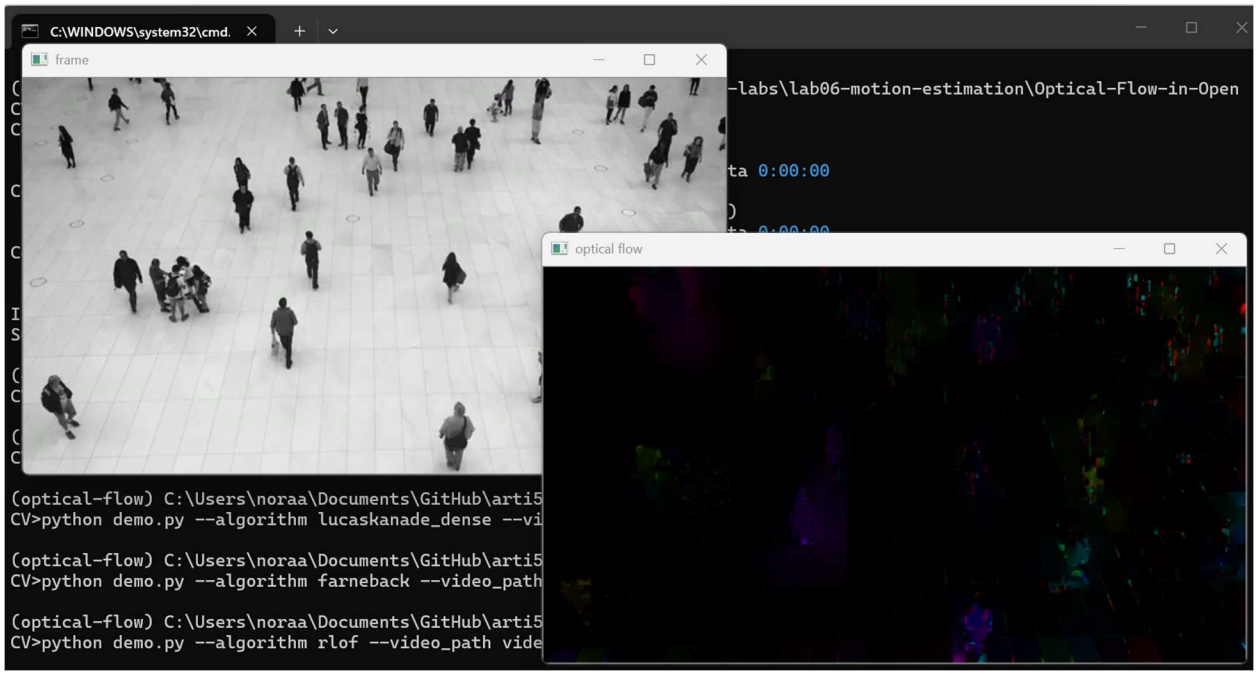
Farneback Optical Flow Result:

```
(optical-flow) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-Open
CV>python demo.py --algorithm farneback --video_path videos/crowd.mp4
```



RLOF Optical Flow Result:

```
(optical-flow) C:\Users\noraa\Documents\GitHub\arti560-computer-vision-labs\lab06-motion-estimation\Optical-Flow-in-Open
CV>python demo.py --algorithm rlof --video_path videos/crowd.mp4
```



In this lab, several optical flow algorithms were applied for motion estimation on the crowd video. Lucas-Kanade was used as a sparse optical flow method to track selected feature points. Dense Lucas-Kanade, Farneback, and RLOF were used to estimate motion more densely across the frames. The results showed differences in how each algorithm represents movement, where sparse methods focus on key points and dense methods provide a fuller motion visualization.